

Class ÷

Class is used to describe properties (data member) and **behaviour** (member function) of an object.

Inheritance ÷

It is a process of inheriting properties and behaviours of existing class into a new class.

Existing class = parent class = base class

New class = child class = Derived class

Syntax ÷ Class base_class

{

};

Class derived_class : **Visibility Mode** base_class

{

};

Types of Inheritance ÷

- Single inheritance
- Multilevel inheritance
- Multiple inheritance
- Hierarchical inheritance
- Hybrid inheritance



Single inheritance :

Class A

{

};

Class B : public A

{

};

A



B

Multi level inheritance :

Class A

{

};

Class B : public A

{

};

Class C : public B

{

};

A



B



C

Multiple inheritance :

Class A1

{

};

Class A2

{

};

Class B : public A1 , public A2

{

};

A1 A2



B

Hierarchical Inheritance :

Class A

{

};

Class B1 : public A

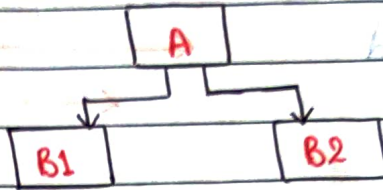
{

};

Class B2 : public A

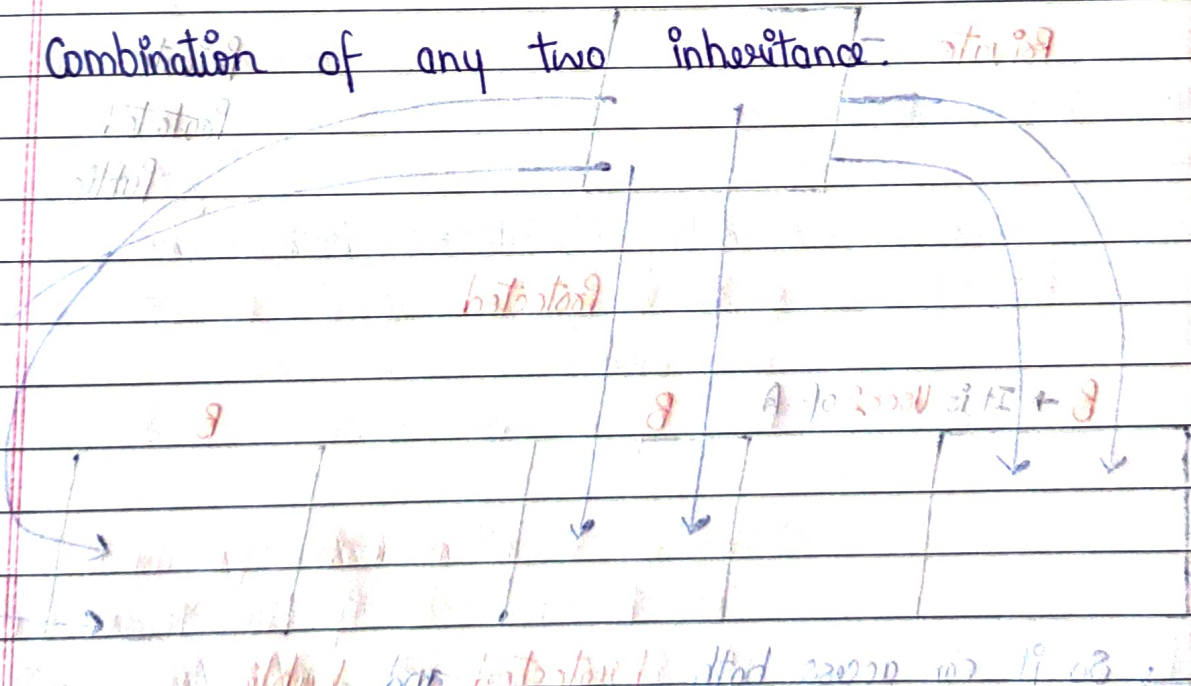
{

};



Hybrid Inheritance :

Combination of any two inheritance.



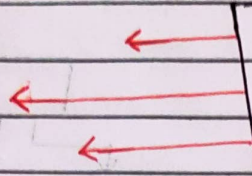
So it can access both of inheritance and hybrid inheritance.
 In hybrid inheritance, the class A has the inheritance and the class B has the inheritance.
 So the hybrid of inheritance is for the class of B.



Visibility Modes ÷

- Private
- Protected
- Public

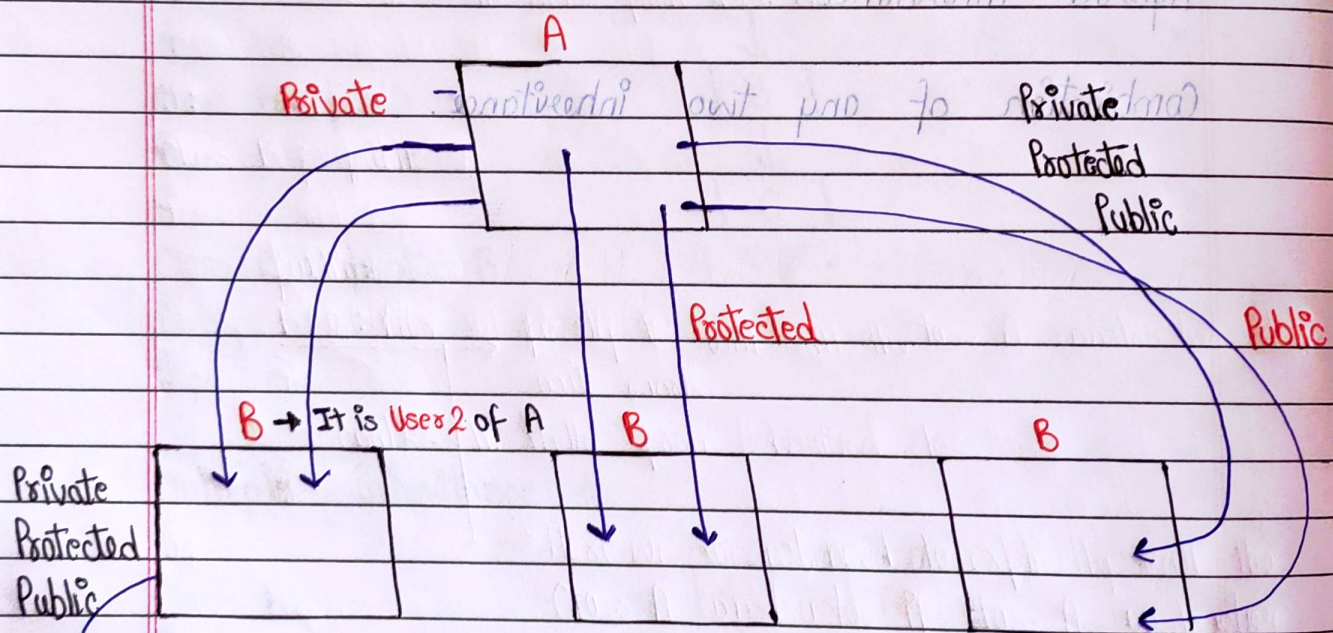
Class A



Types of User ÷

1. User 1 will create object of your class & can only access to **Public** members.
2. User 2 will create **derived class** from your class.

Visibility Modes ÷



→ So it can access both **protected** and **public**

members of **Class A**. Here the **Protected** and **Public** members of **Class A** is **Private** members of **Class B**. So both the users of **Class B** can't access any members of **A**.

*** So the **impact** of **Visibility Mode** is for the Users of **B**
User 1 & User 2



Program:

```
#include <iostream>
```

```
using namespace std;
```

```
class A
```

```
{
```

```
private:
```

```
int a;
```

```
protected:
```

```
void setValue(int k)
```

```
{
```

```
    a = k;
```

```
}
```

```
}
```

```
class B : public A
```

```
{
```

```
public:
```

```
void setData(int x)
```

```
{
```

```
    setValue(x);
```

```
}
```

```
}
```

```
int main()
```

```
{
```

```
    B obj;
```

```
    obj.setData(4);
```

```
}
```